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Maths for Machine Learning

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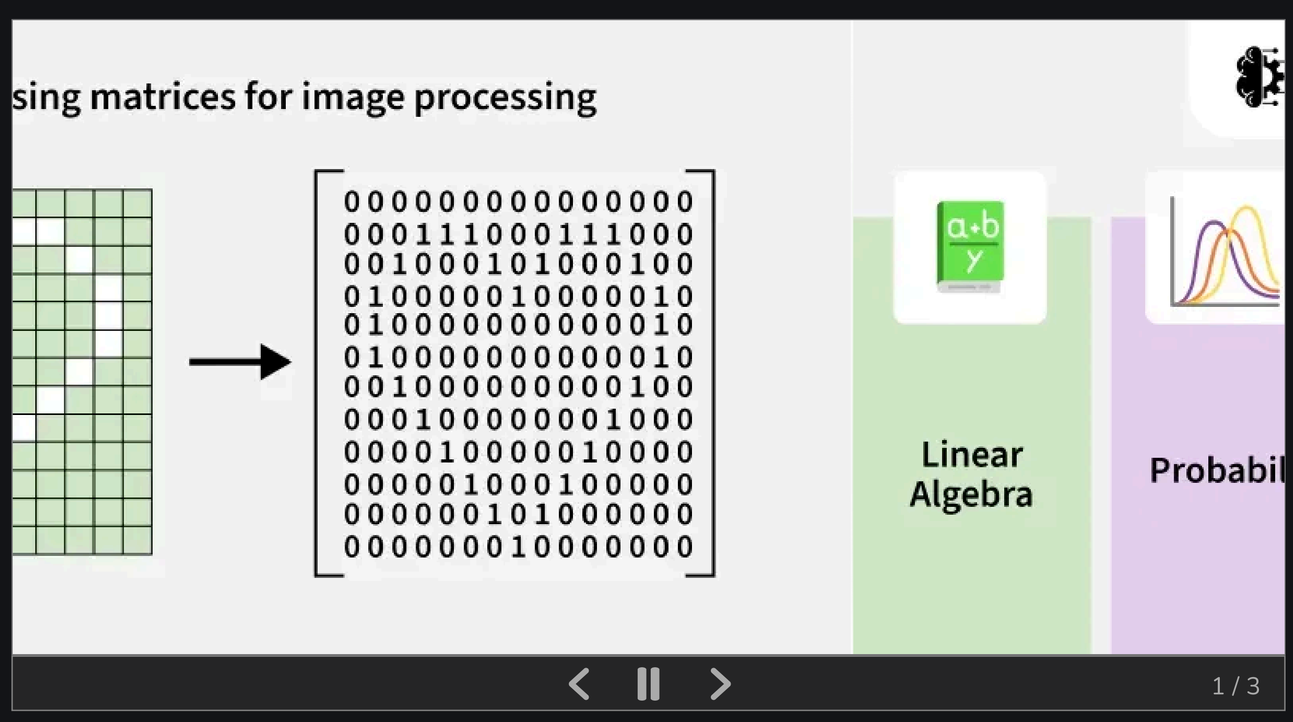
Mathematics is the foundation of machine learning. Math concepts play an important role in understanding how models learn from data and optimizing their performance. They form the base for most machine learning algorithms.

- Builds understanding of data representation and transformation
- Helps in training and optimizing algorithms
- Supports decision-making under uncertainty



Why Learn Mathematics for Machine Learning?

- Math provides the theoretical foundation for understanding how machine learning algorithms work.
- Concepts like calculus and linear algebra enable fine-tuning of models for better performance.
- Knowing the math helps troubleshoot issues in models and algorithms.
- Topics like deep learning, NLP and reinforcement learning require strong mathematical foundations.

How Much Math is Required for Machine Learning?

The amount of math required for machine learning depends on your goals. Let's see the breakdown based on different level:

Basic Understanding (Entry-Level)

- **Linear Algebra:** Basics of vectors, matrices and matrix operations, vector norms, Euclidean distance, Manhattan distance.
- **Statistics:** Descriptive statistics (mean, variance, standard deviation), correlation and covariance, methods of measurement of correlation.
- **Probability:** Basics of probability theory, joint/conditional/marginal probability, Bayes' theorem.
- **Calculus:** Fundamental Calculus Concepts , gradient, Partial Derivatives, Higher-Order Derivatives.

Intermediate Understanding (Practical Implementation)

- **Linear Algebra**: Eigenvalues and Eigenvectors, LU Decomposition, Singular Value Decomposition (SVD)
- **Probability and Statistics**: Central Limit Theorem, Discrete Probability Distributions, Continuous Probability Distributions, hypothesis testing and confidence intervals.
- **Calculus**: Partial Derivatives and chain rule for backpropagation in neural networks.
- **Optimization**: Understanding gradient descent and its variations (e.g., stochastic gradient descent).

Advanced Understanding (Research and Custom Algorithms)

- **Vector Calculus**: Jacobian, Hessian Matrices.
- **Probability Distributions and Statistics**: Sampling Distributions, Chi-Square Distribution, t - Distribution, Parametric Methods, Non-Parametric Test, Bias Vs Variance and Bootstrap method.
- **Geometry:** Cosine Similarity, Jaccard Similarity and Orthogonality and Projections.
- **Regression Analysis:** Maximum Likelihood Estimation (MLE), Mean Squared Error.

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